Introduction

Title:
AI-Driven Fake News Detection Using Rule-Based Reasoning

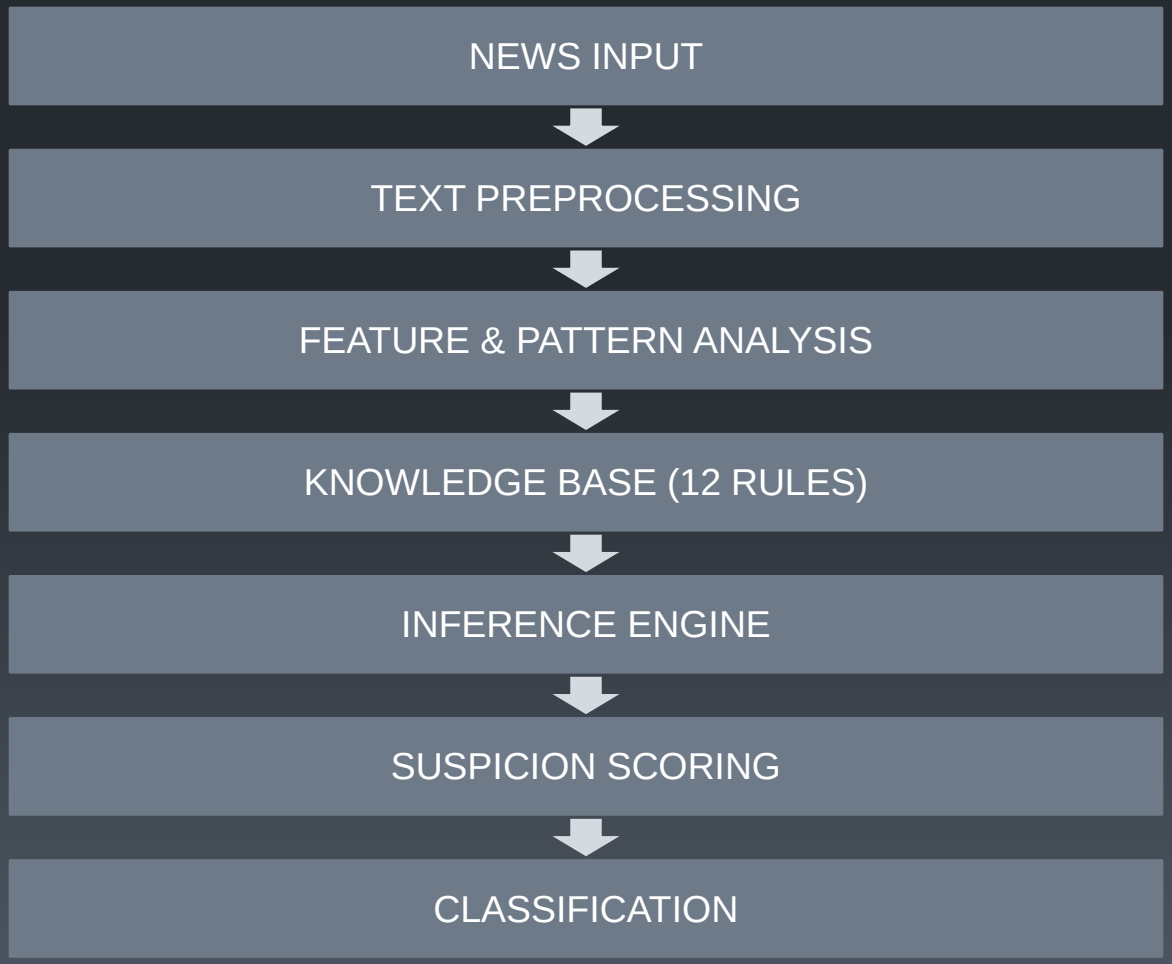
- Fake news spreads rapidly through digital platforms.
- Identifying unreliable information is challenging.
- Uses Rule-Based Artificial Intelligence instead of Machine Learning.
- Provides transparent and explainable decisions.
- Now extended with a 12-rule Knowledge Base for finer-grained scoring.

Existing vs Proposed System

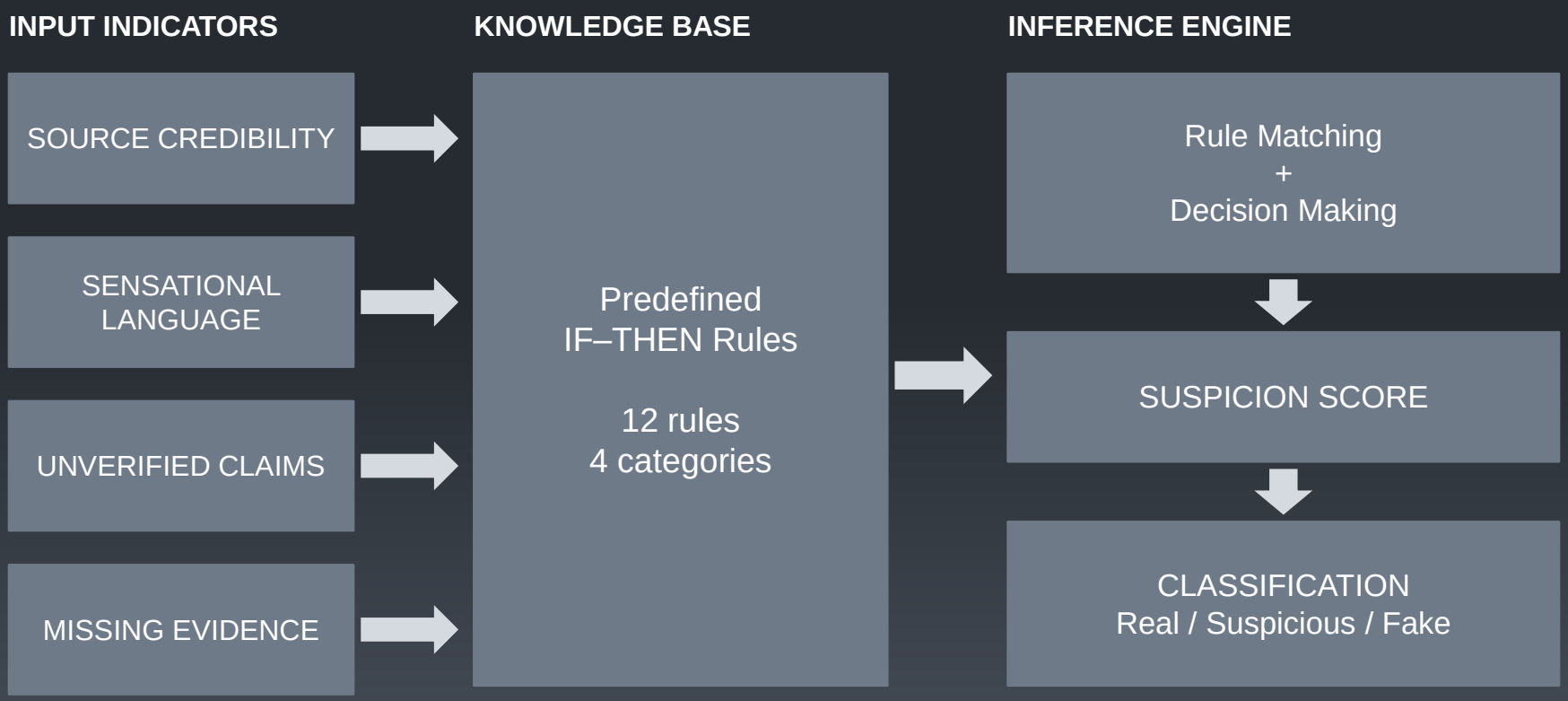
Existing System	Proposed System
Machine Learning / Deep Learning	Rule-Based AI
Requires large datasets	Uses predefined rules
Higher computational requirements	Lightweight
Decisions may be difficult to interpret	Explainable decisions
Requires model training	No ML training

Updated model: the Knowledge Base now holds 12 IF–THEN rules

System Architecture



KNOWLEDGE BASE & RULE-BASED REASONING



Each of the four indicator groups is covered by three rules; triggered rules add points to one suspicion score.

Updated Rule Base: 12 Rules

SOURCE CREDIBILITY

R1	Source & Language	+3
R3	Blacklisted Source	+3
R4	No Author	+1

SENSATIONAL LANGUAGE

R5	Clickbait Phrases	+2
R6	Caps & Punctuation	+1
R7	Emotional Trigger	+2

UNVERIFIED CLAIMS

R2	Unsupported Claim	+2
R8	Absolute Claim	+2
R9	Anonymous Sourcing	+1

MISSING EVIDENCE

R10	No Citations	+2
R11	No Specifics	+1
R12	Fact Conflict	+3

Rule Details: R1 – R6

#	Rule	IF condition	Score	Reason
R1	**Source & Language**	IF source = unknown AND sensational language = high	+3	Unknown source plus hype raises suspicion
R2	**Unsupported Claim**	IF unsupported claim = detected	+2	Claim has no supporting evidence
R3	**Blacklisted Source**	IF source is on the known-unreliable list	+3	Source has a poor track record
R4	**No Author**	IF author / publisher not stated	+1	Accountability is missing
R5	**Clickbait Phrases**	IF clickbait phrase found (e.g. "you won't believe")	+2	Written to attract clicks, not inform
R6	**Caps & Punctuation**	IF excess ALL-CAPS words or 3+ "!" marks	+1	Emotional emphasis over facts

Rule Details: R7 – R12

#	Rule	IF condition	Score	Reason
R7	**Emotional Trigger**	IF fear / outrage word count = high	+2	Designed to provoke, not explain
R8	**Absolute Claim**	IF absolute statement ("proven", "100%") AND no evidence	+2	Overconfident, unbacked claim
R9	**Anonymous Sourcing**	IF only "experts say" / "sources claim" used	+1	Claim cannot be traced or checked
R10	**No Citations**	IF no links or references present	+2	Nothing to verify against
R11	**No Specifics**	IF no dates, figures or named entities	+1	Vague content is hard to verify
R12	**Fact Conflict**	IF statement contradicts a trusted fact in KB	+3	Directly conflicts with known facts

RULE-BASED DECISION EXAMPLES

News → Analysis → Knowledge → Inference → Result

Rule 1: SOURCE & LANGUAGE

IF source = unknown
AND sensational language = high

THEN suspicion score += 3

Reason: Increases suspicion

Rule 2: UNSUPPORTED CLAIM

IF unsupported claim = detected

THEN suspicion score += 2

Reason: Missing support

Scoring & Classification

Suspicion score = sum of the points of every rule that fires (maximum 23 across the 12 rules)

0 – 3	4 – 7	8 +
REAL	**SUSPICIOUS**	**FAKE**
Few or no rules fired. Content looks credible.	Several warning signs. Needs manual verification.	Many rules fired. Strong signs of misinformation.

Worked Example

NEWS INPUT

"SHOCKING!!! Doctors HATE this one trick that cures everything!"

Source: unknown blog. No links, no author.

Rule fired	Points
R1 Source & Language	+3
R5 Clickbait Phrases	+2
R6 Caps & Punctuation	+1
R10 No Citations	+2
Total suspicion score	**8**

Score 8 → FAKE

Every point traces back to a named rule, so the decision is fully explainable.

Conclusion & Future Scope

CONCLUSION:

- Uses Rule-Based AI for news analysis.
- Uses a Knowledge Base + Inference Engine.
- Knowledge Base expanded to 12 IF–THEN rules in 4 categories.
- Produces explainable decisions using predefined rules.
- Does not require Machine Learning training.

FUTURE SCOPE:

- Expand the Knowledge Base beyond 12 rules.
- Add more linguistic rules.
- Integrate external verification sources.
- Add multilingual support.
- Develop a web-based interface.
- Explore hybrid AI approaches.